Performance Testing

Date: 14 July 2026 | **Team ID:** | **Project Name:** Emotion Detection and Learning Support Engine

Performance testing was conducted to evaluate the responsiveness, stability, and resource utilization of the Emotion Detection and Learning Support Engine under normal operating conditions.

Step 1: Testing Overview

Field	Details
Testing Tools Used	Streamlit Application, Windows Task Manager, Manual Functional Testing
Type of Testing	Performance Testing (Response Time & Resource Utilization)
Target Module / API	Emotion Detection Module, Learning Guidance Module, Analytics Dashboard
Test Environment	Windows 11, Python 3.11, Streamlit, TensorFlow, Intel Core i5, 8 GB RAM
Test Date	14 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration	Expected Outcome
1	Launch the application	1	1 minute	Application starts successfully.
2	Predict emotion from text input	1	2 minutes	Emotion prediction displayed within a few seconds.
3	Generate learning guidance	1	1 minute	Personalized guidance displayed correctly.
4	Display analytics dashboard	1	1 minute	Dashboard loads without errors.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 3 sec	1.8 sec	Pass	Fast response
2	Response Time (Max)	< 5 sec	3.4 sec	Pass	Within range
3	Throughput	≥ 15 req/min	18 req/min	Pass	Satisfactory
4	Error Rate	0%	0%	Pass	No errors
5	CPU Utilization	< 70%	42%	Pass	Efficient
6	Memory Utilization	< 80%	55%	Pass	Stable

Step 4: Observations & Analysis

Key Findings: The application responds quickly to user inputs, emotion prediction is generated successfully without noticeable delays, analytics dashboard loads correctly, and the system remains stable.

Bottleneck Identified: Initial application startup takes slightly longer due to loading the trained BiLSTM model into memory.

Optimization Steps Taken: Optimized model loading, modular component organization, efficient CSV logging, and reuse of pre-trained tokenizer/label encoder.

Step 5: Screenshots / Evidence